

ELECTRICAL NETWORK ANALYSIS AND SYNTHESIS LAB

Name: - Dheerendra Amrute

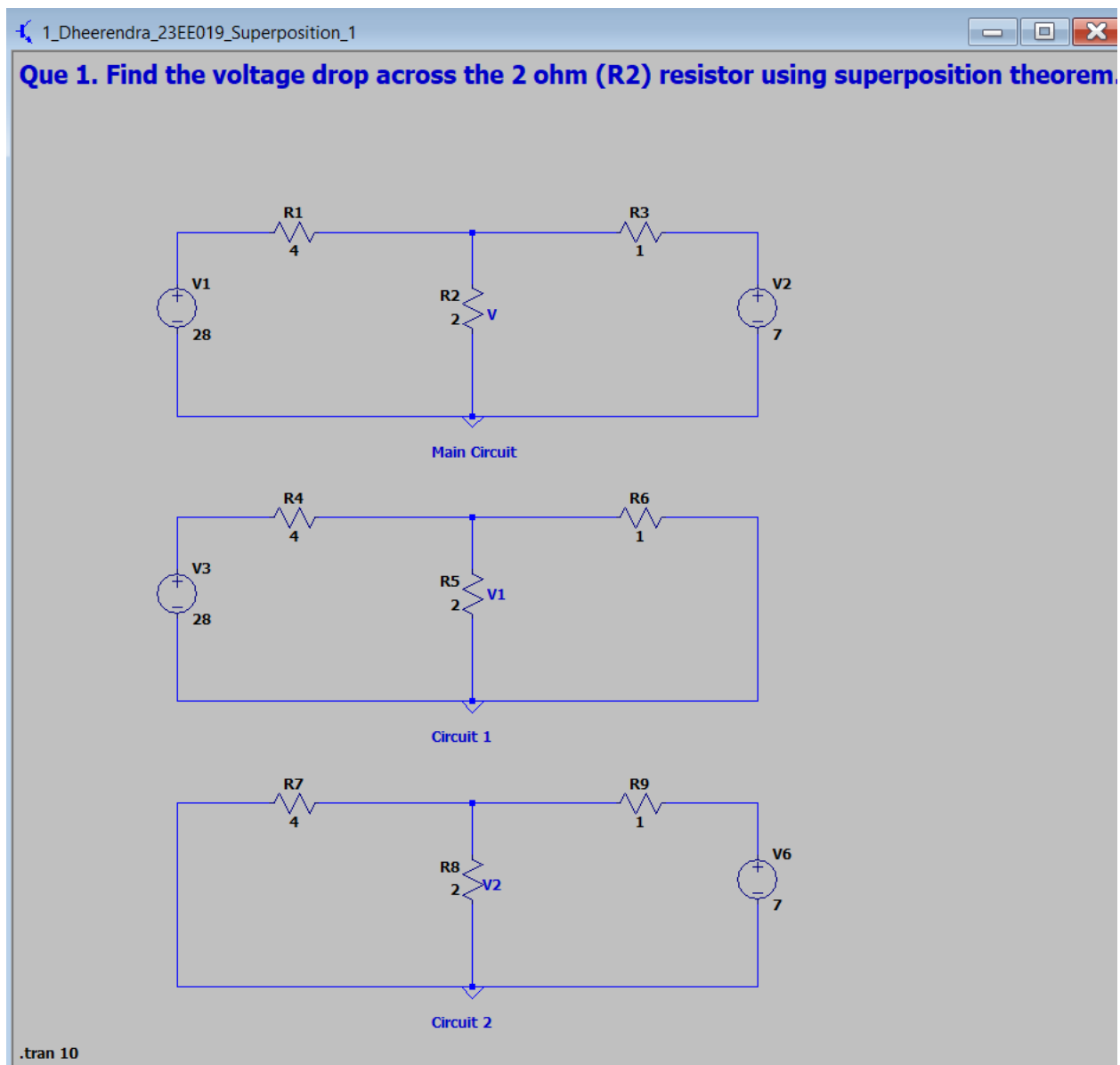
Roll No: - 23EE019

LAB – 1

Experiment 1 Superposition Theorem

Question 1: Find the voltage drop across the 2-ohm (R_2) resistor using superposition theorem.

Circuits: -



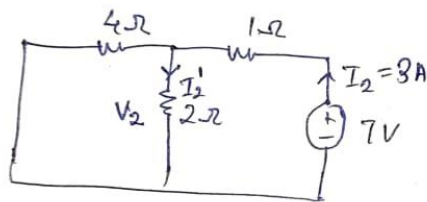
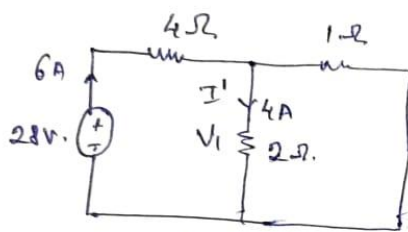
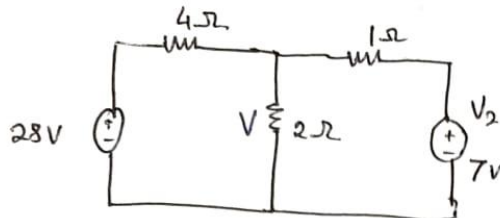
Response: -



Theoretical Calculations: -

Electrical Network LAB

Que 3 |



$$V = V_1 + V_2$$

$$V = 4 + 4 = 8V$$

$$\boxed{V = 8V}$$

$$V = V_1 + V_2$$

For V1 :-

$$R_{eq} = \frac{2 \times 1}{2+1} + 4$$

$$R_{eq} = \frac{2}{3} + 4 \Rightarrow \frac{2+12}{3} = \frac{14}{3}$$

$$I = \frac{28 \times 3}{14} = 6A$$

$$I' = \frac{4 \times 6^2}{8} = 2A, \quad V_1 = I' \times 2 = 2 \times 2 = 4V$$

$$\boxed{V_1 = 4V}$$

For V2 :-

$$R_{eq} = \frac{4 \times 2}{6} + 1 \Rightarrow \frac{8+6}{6} = \frac{14}{6}$$

$$I_2 = \frac{28 \times 6^3}{14 \times 2} = 3A$$

$$I_2' = \frac{4 \times 3^2}{6 \times 2} = 2A$$

$$V_2 = I_2' \times 2$$

$$V_2 = 2 \times 2 = 4V$$

$$\boxed{V_2 = 4V}$$

Observation Table: -

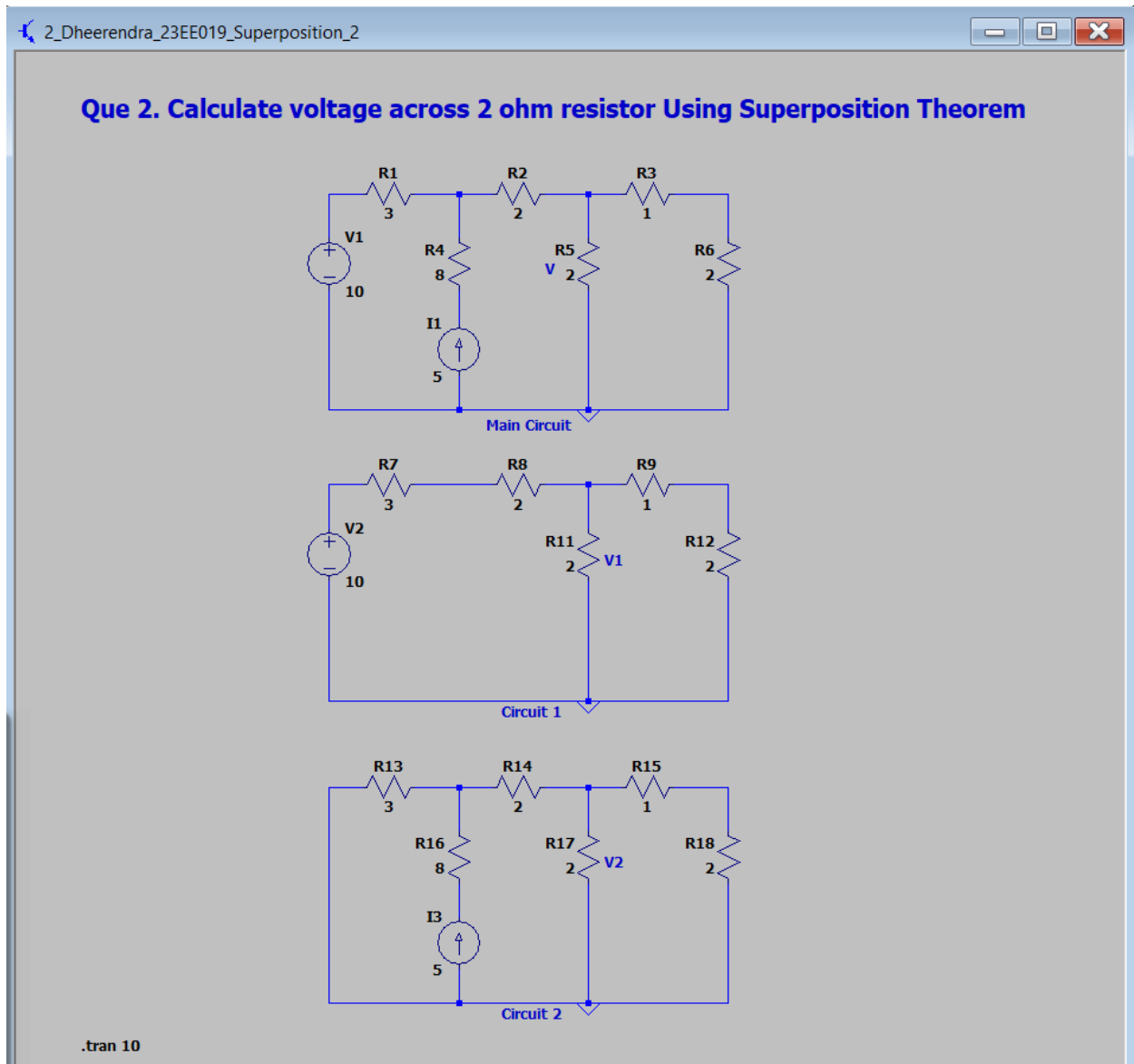
Parameter	Theoretical Values (in volts)	Simulation Result (in volts)
V1	4	4
V2	4	4
V	8	8

Result: -

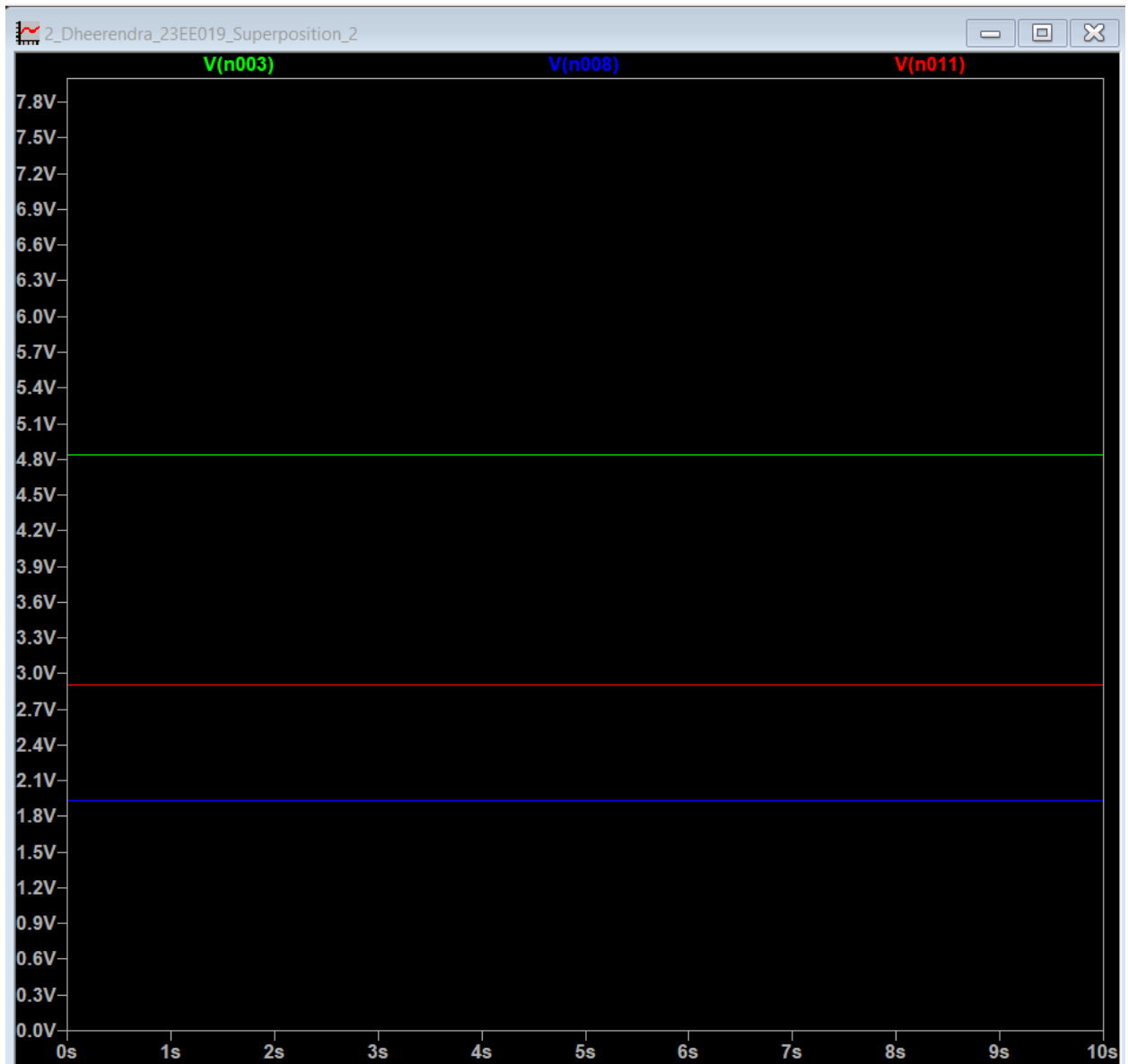
As we observe that the theoretical values are equal to the simulated values hence superposition theorem is verified.

Question 2: Calculate voltage across 2-ohm resistor Using Superposition Theorem.

Circuits: -

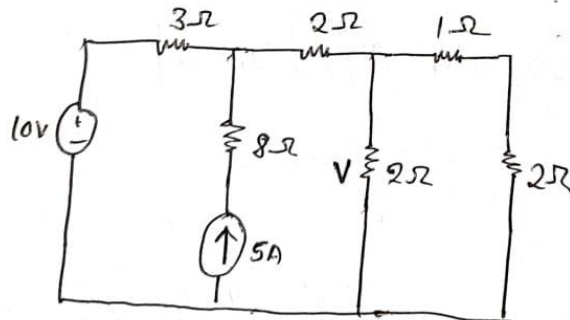


Response: -



Theoretical Calculations: -

Que 2



$$V = V_1 + V_2$$

For V_1 :-

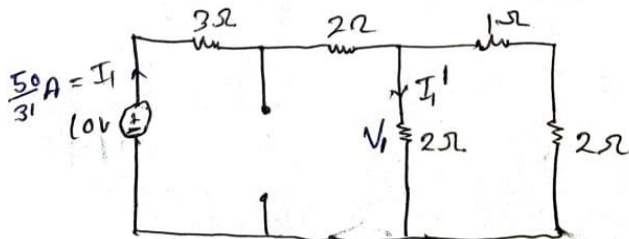
$$R_{eq} = \frac{3 \times 2}{3+2} + 5 = \frac{6}{5} + 5$$

$$= \frac{6+25}{5} = \frac{31}{5} \Omega$$

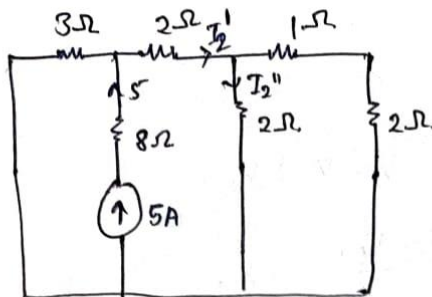
$$I_1 = \frac{10 \times 5}{31} = \frac{50}{31} A$$

$$I_1' = \frac{3 \times \frac{50}{31}}{2+3} = \frac{30}{31} = 0.968$$

$$V_1 = I_1' \times 2 = \frac{30}{31} \times 2 = \frac{60}{31} = \boxed{1.935 A = V_1}$$



10



$$R_{eq} = \frac{3 \times 2}{3+2} + 2 = \frac{6}{5} + 2 = \frac{6+10}{5} = \frac{16}{5} \Omega$$

$$I_2' = \frac{5 \times 3}{\frac{16}{5} + 3} = \frac{15 \times 5}{31}$$

$$I_2' = \frac{75}{31} A$$

$$I_2'' = \frac{3 \times \frac{75}{31}}{5} = 1.4516 A \Rightarrow V_2 = I_2'' \times 2 = \frac{45}{31} \times 2$$

$$\boxed{V_2 = 2.9032 V}$$

$$V = V_1 + V_2 \Rightarrow 2.9032 + 1.935 = 4.838 V$$

$$\boxed{V = 4.838 V}$$

Observation Table: -

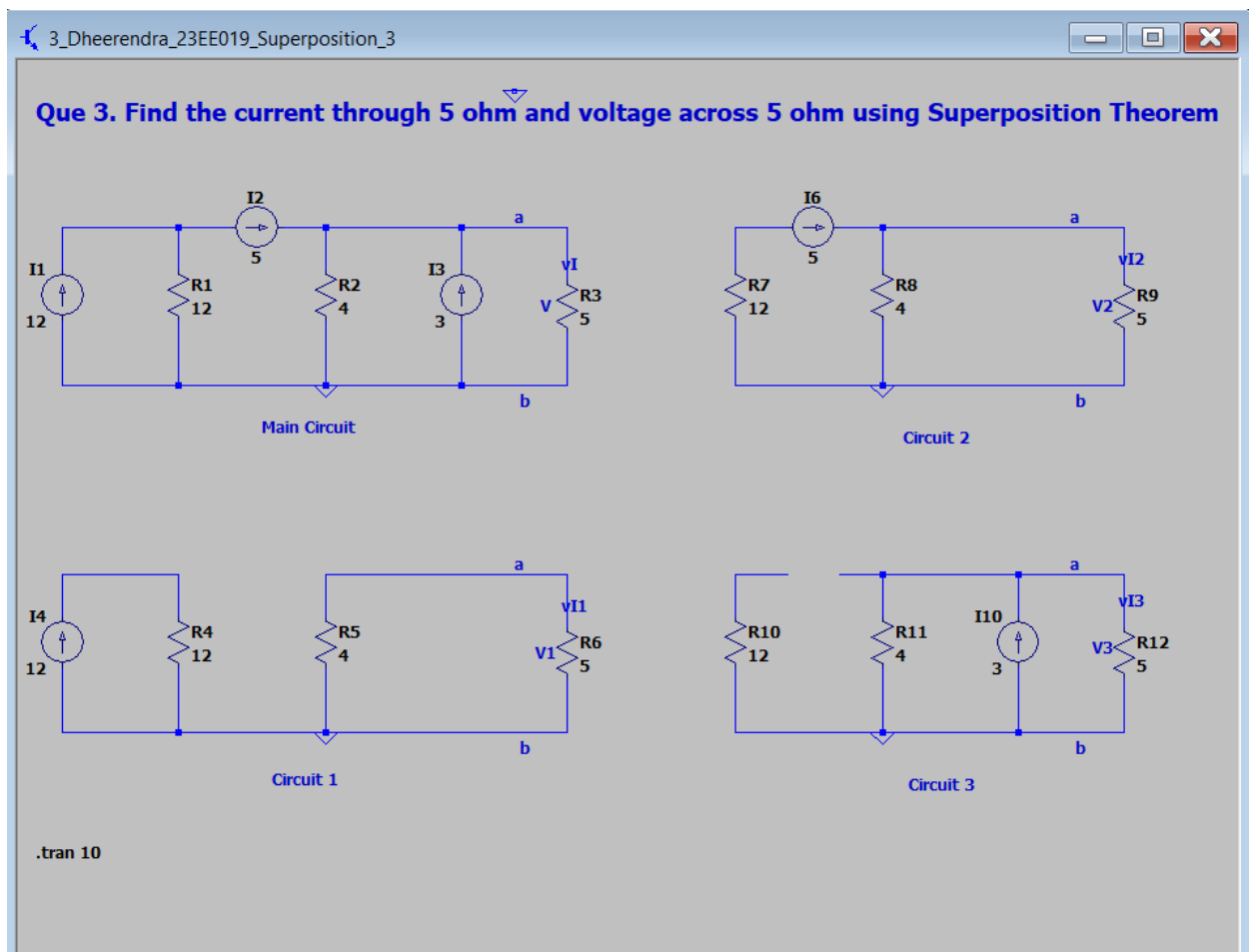
Parameter	Theoretical Values (in volts)	Simulation Result (in volts)
V1	1.935	1.9354838
V2	2.903	2.9032259
V	4.838	4.8387098

Result: -

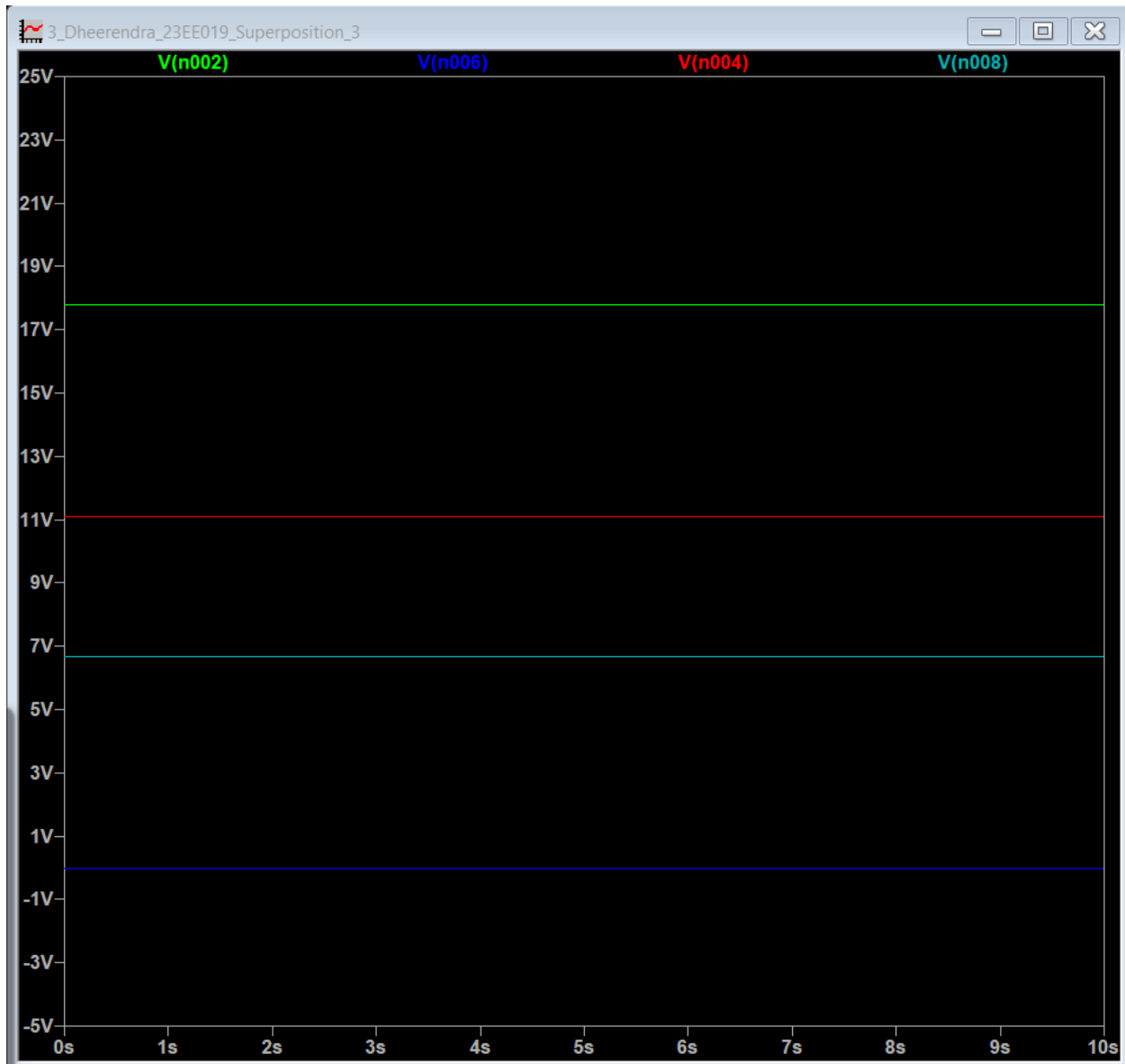
As we observe that the theoretical values are equal to the simulated values hence superposition theorem is verified.

Question 3: Find the current through 5 ohm and voltage across 5 ohm using Superposition Theorem.

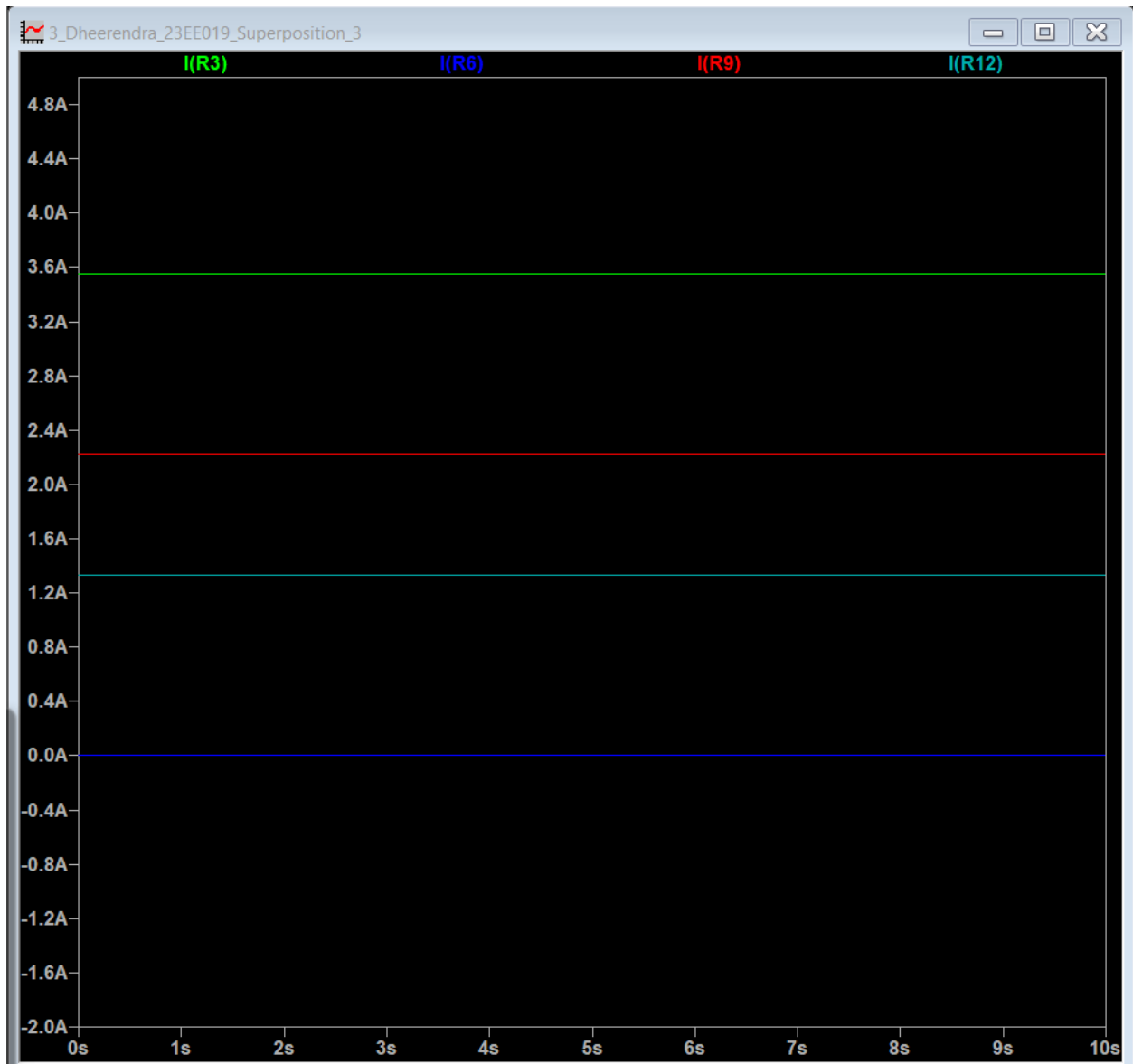
Circuits: -



Response: -



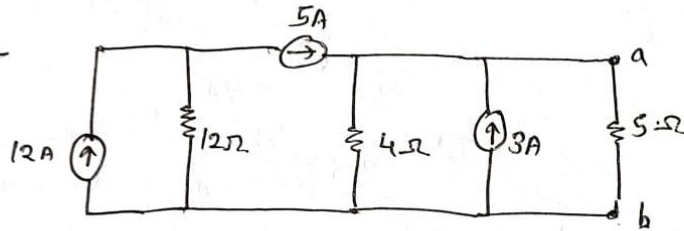
Voltage vs Time



Current vs Time

Theoretical Calculations: -

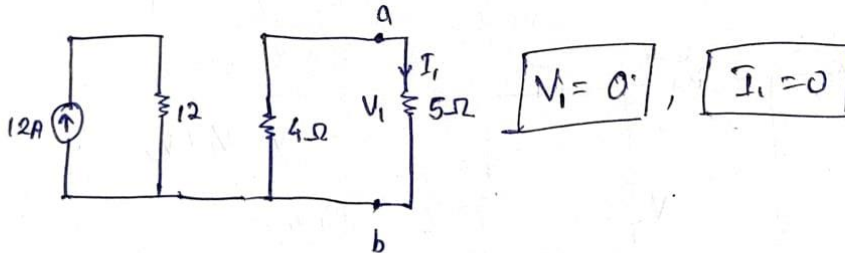
Que 3



$$V = V_1 + V_2 + V_3$$

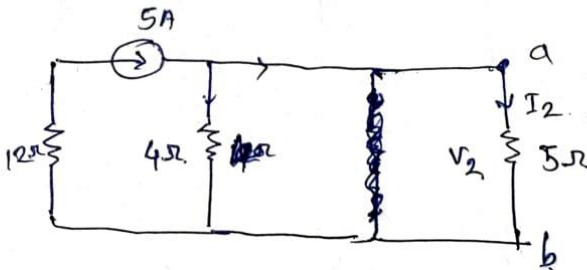
$$I = I_1 + I_2 + I_3$$

①



$$V_1 = 0, \quad I_1 = 0$$

②

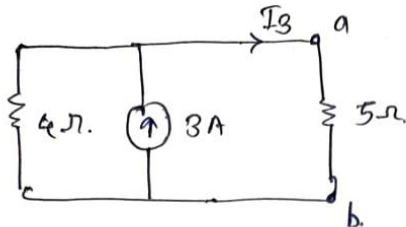


$$I_2 = \frac{4 \times 5}{9} = \frac{20}{9} = 2.2 \text{ A}$$

$$V_2 = I_2 \times 5 = 2.22 \times 5 = 11.1 \text{ V}$$

$$V_2 = 11.1 \text{ V}$$

$$I_2 = 2.2 \text{ A}$$



$$I_3 = \frac{4 \times 3}{8} = \frac{4}{3}$$

$$I_3 = 1.33 \text{ A}$$

$$V_3 = I_3 \times 5 = \frac{4}{3} \times 5 = 6.667 \text{ V}$$

$$V_3 = 6.67 \text{ V}$$

$$V = V_1 + V_2 + V_3$$

$$= 0 + 11.1 + 6.67 = 17.77$$

$$V = 17.77 \text{ V}$$

$$I = I_1 + I_2 + I_3$$

$$I = 0 + 2.2 + 1.33$$

$$I = 3.53 \text{ A}$$

Observation Table (for Voltage): -

Parameter	Theoretical Values (in volts)	Simulation Result (in volts)
V1	0	0
V2	11.11	11.111111
V3	6.67	6.666665
V	17.77	17.777779

Observation Table (for Current): -

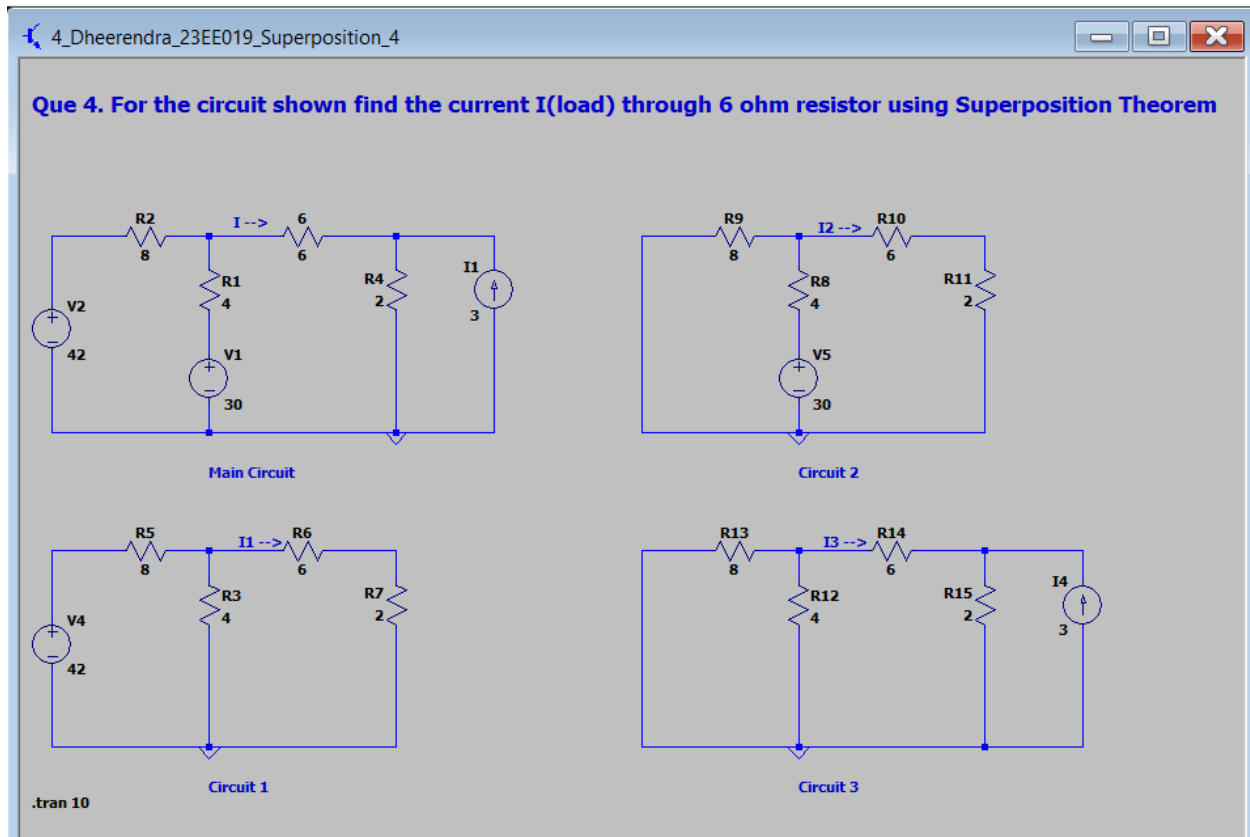
Parameter	Theoretical Values (in amp)	Simulation Result (in amp)
I1	0	0
I2	2.22	2.222223
I3	1.33	1.333334
I	3.55	3.555556

Result: -

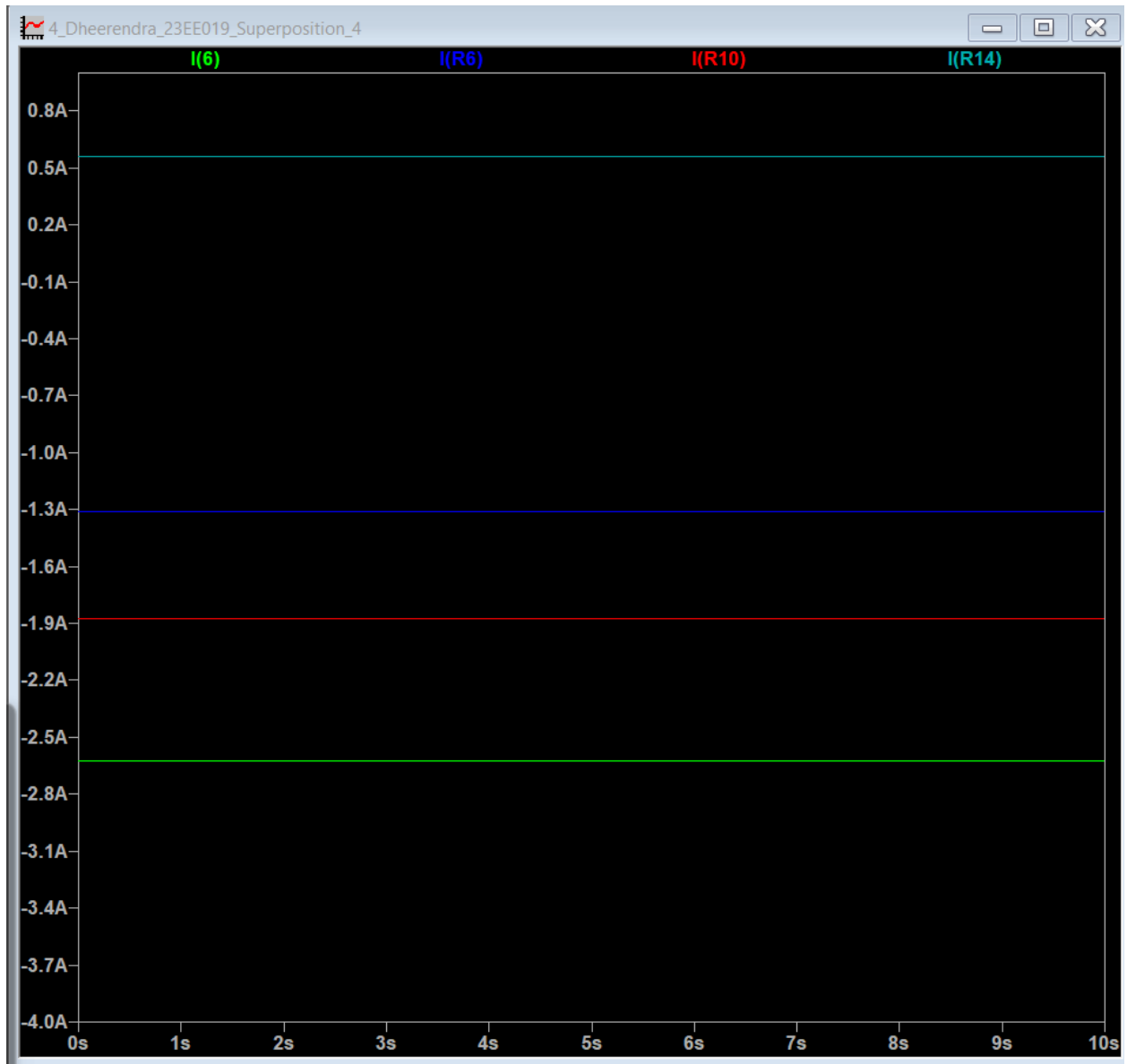
As we observe that the theoretical values are equal to the simulated values hence superposition theorem is verified.

Question 4: For the circuit shown find the current $I(\text{load})$ through 6-ohm resistor using Superposition Theorem.

Circuits: -

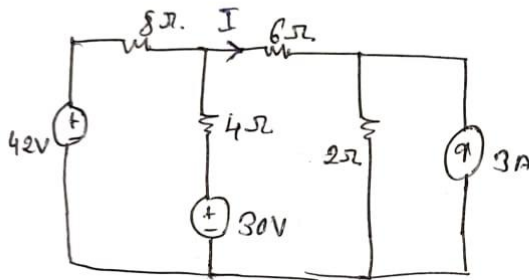


Response: -



Theoretical Calculations: -

Ques 4



$$V = V_1 + V_2 + V_3$$

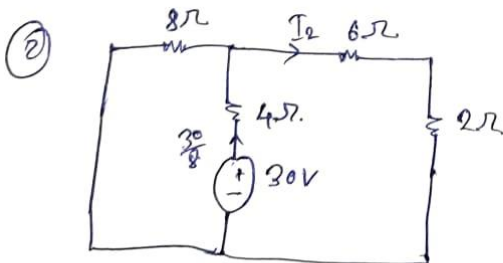
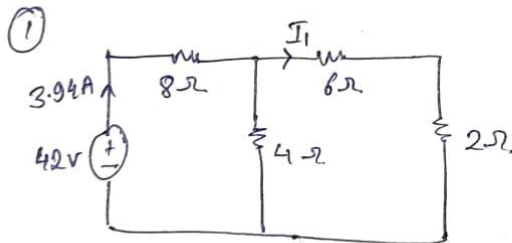
$$I = I_1 + I_2 + I_3$$

For I_1 :-

$$R_{eq} = \frac{8 \times 4}{8+4} + 8 = 10.667 \Omega$$

$$I_1 = \frac{4 \times 3.94}{4+8} = 1.313 A$$

$$\boxed{I_1 = 1.31 A}$$

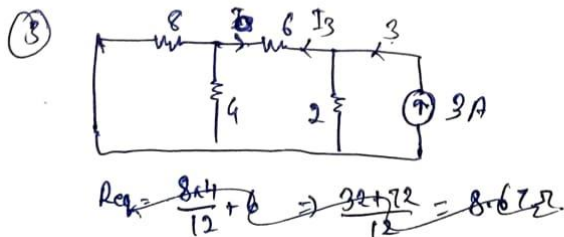


For I_2 :-

$$R_{eq} = 8 \Omega$$

$$\boxed{I_2 = 1.875}$$

$$I_2 = \frac{30}{8+8} = \frac{30}{16} = 1.875$$



$$I_3 = \frac{2 \times 3}{8.67+2} = 0.56$$

$$\boxed{I_3 = -0.56 A}$$

$$I = I_1 + I_2 + I_3 \Rightarrow 1.31 + 1.875 - 0.56$$

$$\boxed{I = 2.625 A}$$

Observation Table (for Current): -

Parameter	Theoretical Values (in amp)	Simulation Result (in amp)
-----------	-----------------------------	----------------------------

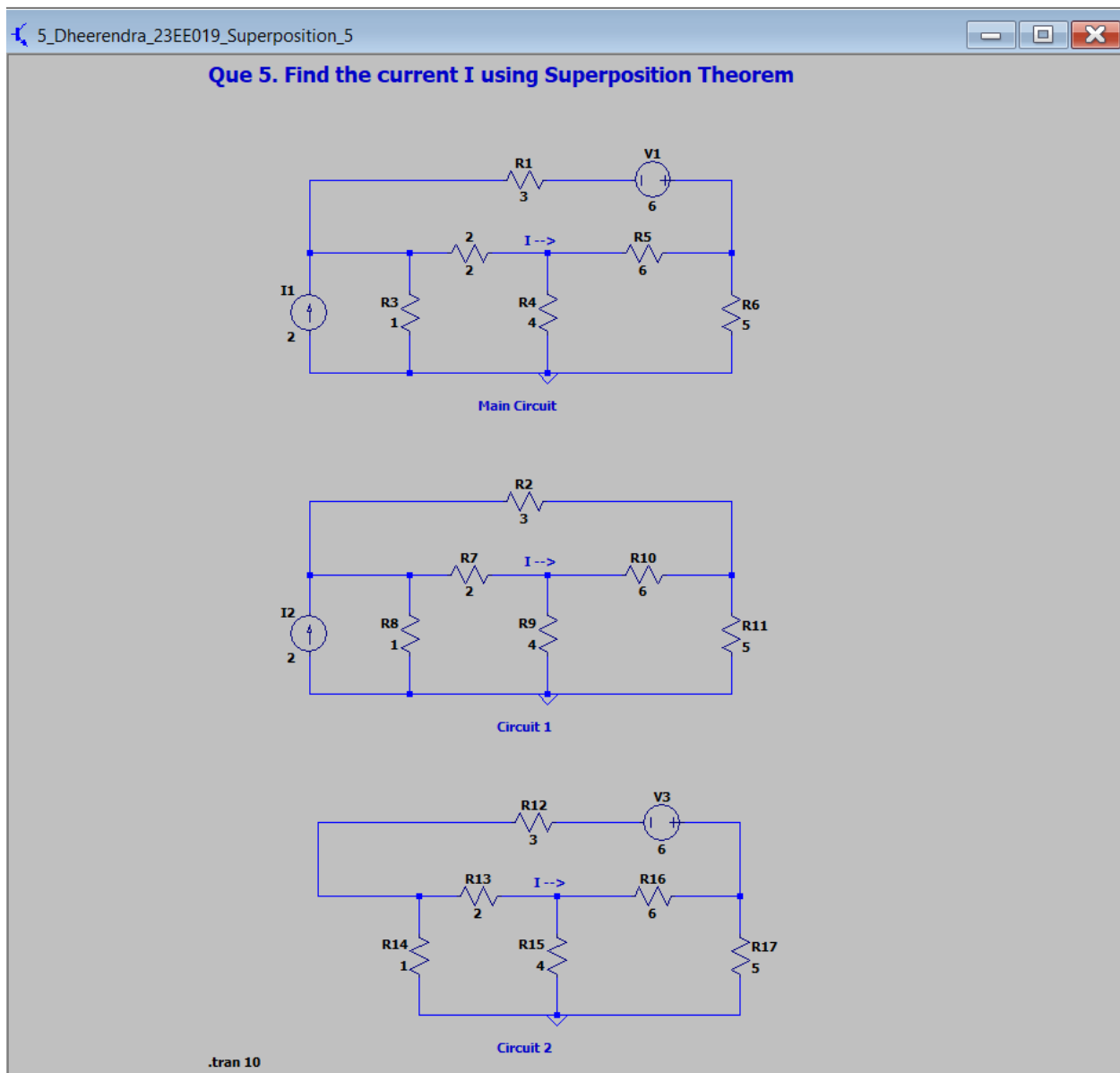
I1	-1.31	-1.3125
I2	-1.875	-1.875
I3	0.56	0.5625
I	-2.625	-2.625

Result: -

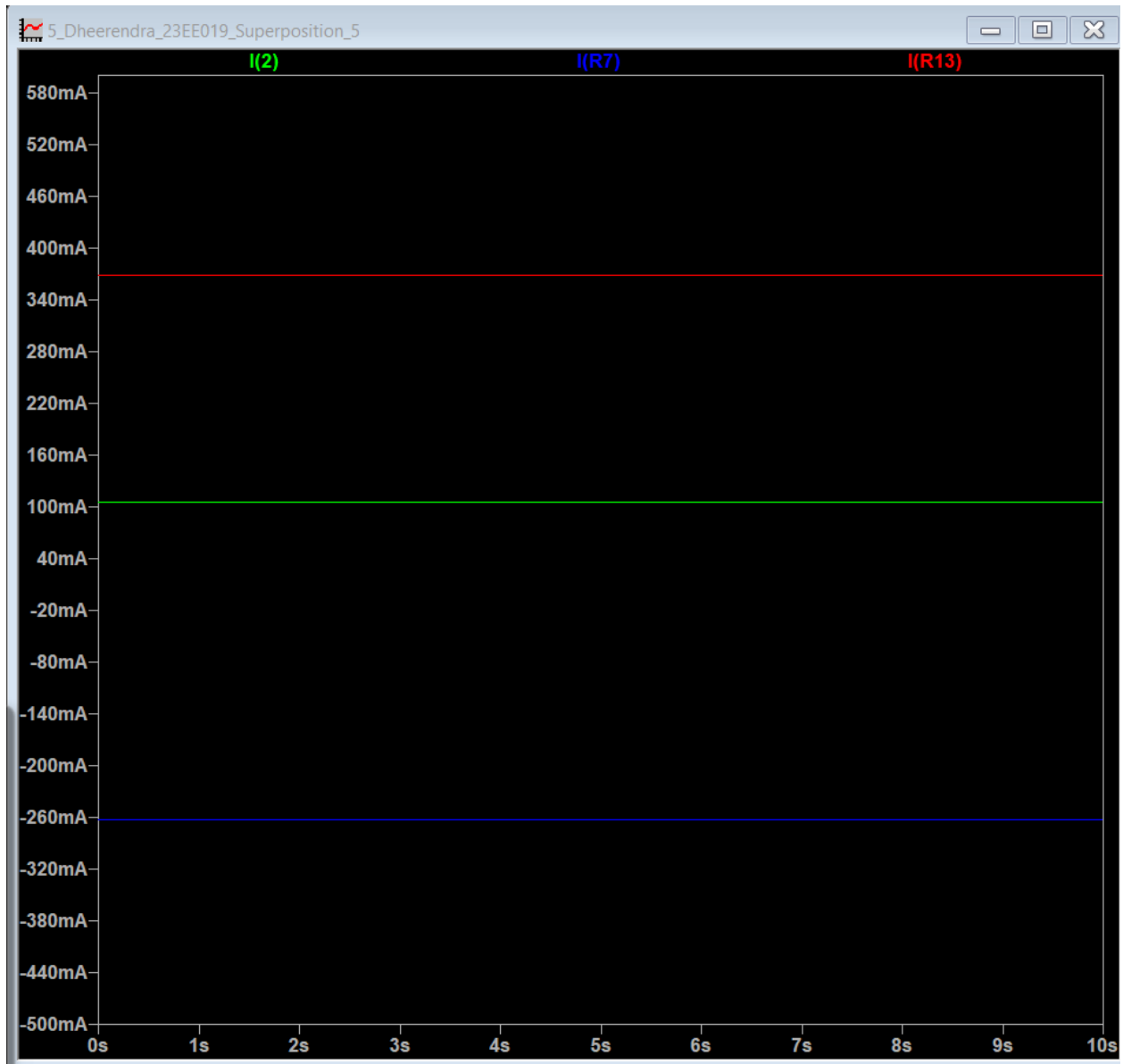
As we observe that the theoretical values are equal to the simulated values hence superposition theorem is verified.

Question 5: Find the current I using Superposition Theorem.

Circuits: -

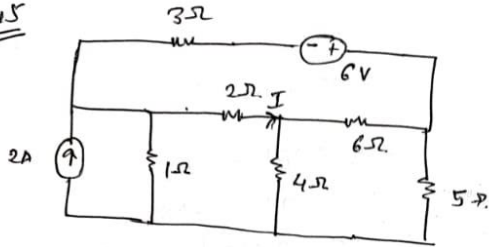


Response: -

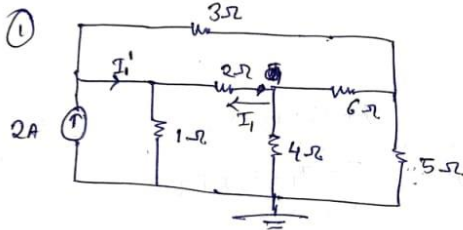


Theoretical Calculations: -

Ques 5



$$I = I_1 + I_2$$



$$R_{eq} = \frac{11 \times 4}{11+4} + 2 = \frac{74}{15} = 4.933$$

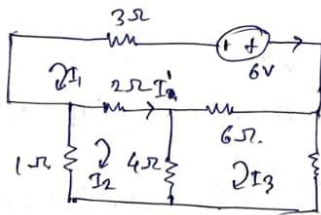
$$R_{eq} = \frac{4.93 \times 1}{4.93+1} = \frac{4.93}{5.933}$$

$$R_{eq} = 0.831 \Omega$$

$$I_1 = \frac{3 \times 2}{3.831} = 1.566 A$$

$$I_1 = \frac{1 \times 1.566}{1 + 4.933} = +0.264 A$$

②



For I_2

$$\Rightarrow 6 - 6(I_1 - I_3) - 2(I_1 - I_2) - 3I_1 = 0$$

$$\Rightarrow -6(I_3 - I_1) - 5I_3 - 4(I_3 - I_2) = 0$$

$$\Rightarrow -2(I_2 - I_1) - 4(I_2 - I_3) - I(I_2) = 0$$

$$\text{Now, } 6 - 11I_1 + 6I_3 + 2I_2 = 0$$

$$-15I_3 + 6I_1 + 4I_2 = 0$$

$$-7I_2 + 2I_1 + 4I_3 = 0$$

$$I_1 = \frac{534}{571}, I_2 = \frac{324}{571}, I_3 = \frac{300}{571}$$

$$= 0.935 A$$

$$= 0.567$$

$$= 0.525$$

$$I' = I_1 + I_2 \Rightarrow \frac{534 - 324}{571} \Rightarrow \underline{\underline{0.367 A}}$$

$$I' = I_2 = 0.367$$

$$I = -I_1 + I_2 \Rightarrow -0.264 + 0.367 = \underline{\underline{+0.103 A}}$$

Observation Table (for Current): -

Parameter	Theoretical Values (in amp)	Simulation Result (in amp)
I1	-0.264	-0.26269701
I2	0.367	0.36777583
I	0.103	0.10507881

Result: -

As we observe that the theoretical values are equal to the simulated values hence superposition theorem is verified.